Numerical Analysis I

Quiz 7B Name:_____

Consider the four data points

x	-2	-1	1	2
$f(x)$	0	3	-3	0

Using the **Newton divided-difference algorithm**:

1. Construct the complete divided-difference table.
2. Write the degree-3 interpolating polynomial in **nested Newton form**.
3. Use your polynomial to evaluate $P_3(3)$.

Definition. Divided differences Let $f : \mathbb{R} \to \mathbb{R}$. Let $x_0, x_1, \ldots, x_n$ be $n+1$ distinct points. Define $f[x_0, x_1, \ldots, x_n]$.

Theorem. Recursive formula for divided differences Let $x_0, x_1, \ldots, x_n$ be distinct points. Then

$$\boxed{f[x_0, x_1, \ldots, x_n] = \frac{f[x_1, x_2, \ldots, x_n] - f[x_0, x_1, \ldots, x_{n-1}]}{x_n - x_0}.}$$

Let P_n be the unique polynomial of degree at most n that interpolates f at the $n+1$ points:

$$x_0, x_1, \ldots, x_n.$$

In Newton form,

$$P_n(x) = \underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}$$

$$\underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad} \quad (1)$$

The coefficient of x^n in P_n is

$$\underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad} \quad (2)$$

Now let Q_{n-1} be the interpolation polynomial of degree at most $n-1$ through the n data points:

$$(x_0, f(x_0)), \ldots, (x_{n-1}, f(x_{n-1})),$$

The coefficient of x^{n-1} in Q_{n-1} is

$$\underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad} \quad (3)$$

And let R_{n-1} be the interpolation polynomial of degree at most $n-1$ through the n data points:

$$(x_1, f(x_1)), \ldots, (x_n, f(x_n)).$$

The coefficient of x^{n-1} in R_{n-1} is

$$\underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad} \quad (4)$$

Consider the polynomial

$$S_n(x) = \frac{(x - x_0)R_{n-1}(x) - (x - x_n)Q_{n-1}(x)}{x_n - x_0}.$$

We claim that $S_n = P_n$.

First the degree of S_n is $\underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}$. (5)

because $\underline{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}$. (6)

Second, we claim that S_n interpolates f at the $n+1$ points:

$$x_0, x_1, \ldots, x_n.$$

Indeed, for $i = 1, \ldots, n-1$, we have

$$S_n(x_i) = \underline{\qquad\qquad} \tag{7}$$

$$\underline{\qquad\qquad} \tag{8}$$

At the two endpoints,

$$S_n(x_0) = \underline{\qquad\qquad} \tag{9}$$

$$\text{because } \underline{\qquad\qquad} \tag{10}$$

and

$$S_n(x_n) = \underline{\qquad\qquad} \tag{11}$$

$$\text{because } \underline{\qquad\qquad} \tag{12}$$

Thus S_n interpolates f at all $n+1$ points $x_0, \ldots, x_n$.
S_n and P_n are two polynomials of degree at most n that interpolates f at the $n+1$ points $x_0, \ldots, x_n$, thus by uniqueness of the interpolating polynomial, we have

$$S_n = P_n.$$

It remains to compare the coefficients of x^n. The coefficient of x^n in

$$(x - x_0)R_{n-1}(x)$$

is

$$\underline{\qquad\qquad} \tag{13}$$

While the coefficient of x^n in

$$(x - x_n)Q_{n-1}(x)$$

is

$$\underline{\qquad\qquad} \tag{14}$$

Therefore the coefficient of x^n in S_n is

$$\underline{\qquad\qquad} \tag{15}$$

But $S_n = P_n$, and the coefficient of x^n in P_n is

$$\underline{\qquad\qquad} \tag{16}$$

Consequently, our conclusion is

$$\underline{\qquad\qquad} \tag{17}$$